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10. Convert a velocity of 72 kmh^{-1} into ms^{-1} with the help of dimensional analysis.

Solution:

$n_1 = 72\ kmh^{-1}$ $\quad n_2 = ?\ ms^{-1}$

$L_1 = 1Km$ $\quad L_2 = 1m$

$T_1 = 1h$ $\quad T_2 = 1s$

$$n_2 = n_1\left[\frac{L_1}{L_2}\right]^a\left[\frac{T_1}{T_2}\right]^b$$

The dimensional formula for velocity is $[L\ T^{-1}]$

$a = 1\ b = -1$

$$n_2 = 72\left[\frac{1Km}{1m}\right]^1\left[\frac{1h}{1s}\right]^{-1}$$

$$n_2 = 72\left[\frac{1000m}{1m}\right]^1\left[\frac{3600s}{1s}\right]^{-1}$$

$$= 72 \times 1000 \times 1/3600 = 20 ms^{-1}]$$

$$72\ kmh^{-1} = 20\ ms^{-1}$$

11. Check the correctness of the following equation using dimensional analysis. Make a comment on it.

S = ut + 1/4 at^2 where s is the displacement, u is the initial velocity, t is the time and a is the acceleration produced.

Solution:

Dimension for distance s = [L]

Dimension for initial velocity

$$v = [LT^{-1}]$$

Dimension for time t = [T]

Dimension for acceleration

$$a = [LT^{-2}]$$

According to the principle of homogeneity,

Dimensions of LHS = Dimensions of RHS

Substituting the dimensions in the given formula

S = ut + 1/4 at^2, $\frac{1}{4}$ is a number. It has no dimensions

$$[L] = [LT^{-1}]\ [T^1] + [LT^{-2}]\ [T^2]$$

$$[L] = [L] + [L]$$

As the dimensional formula of LHS is same as that of RHS, the equation is dimensionally correct.

Comment:

But actually it is a ***wrong*** equation. We know that the equation of motion is $s = ut + 1/2\ at^2$

So, a dimensionally correct equation ***need not*** be the true (or) actual equation

But a true equation is ***always*** dimensionally correct.

12. Round - off the following numbers as indicated.

a) 17.234 to 3 digits

b) 3.996×10^5 to 3 digits

c) 3.6925×10^{-3} to 2 digits

d) 124783 to 5 digits.

Solution:

a) 17.2 $\quad$ b) 4.00×10^5

c) 3.7×10^{-3} $\quad$ d) 124780

13. Solve the following with regard to significant figures.

a) $\sqrt{4.5 - 3.31}$

b) $5.9 \times 10^5 - 2.3 \times 10^4$